

Bike Sharing Demand Forecasting: A Comparative Analysis of Machine Learning and Time Series Approaches

Abstract

This report details the analysis of hourly bike-sharing rental data to forecast demand. We explore the dataset using R to understand fundamental relationships and feature interactions. Subsequently, we construct and compare two distinct modeling approaches in Python: a time series approach using SARIMAX and a machine learning regression approach using Histogram-based Gradient Boosting. The report outlines the rationale for model selection, mathematical formulations, and a final comparative evaluation.

1 Introduction

Accurate forecasting of bike-sharing demand is critical for system optimization and re-balancing. The dataset provides hourly rental counts (`cnt`) alongside calendar variables (season, year, month, hour, holiday, weekday, workingday) and meteorological variables (weather situation, temperature, feeling temperature, humidity, windspeed). The objective is to leverage these exogenous factors and historical rental patterns to accurately predict future hourly demand.

2 Exploratory Data Analysis (EDA) and Feature Engineering

Initial exploratory analysis and baseline modeling were conducted in R. The baseline approach utilized Multiple Linear Regression (MLR) to establish the primary drivers of demand.

Key insights and engineered features from the EDA phase included:

- **Non-linear Weather Effects:** Temperature and humidity exhibit non-linear relationships with bike rentals. To capture this, squared terms (`temp2` and `hum2`) were engineered.
- **Categorical Handling of Time:** Treating the hour of the day (`hr`) as a continuous variable is flawed due to the cyclical nature of rush hours. Converting `hr` to a factor (`hr_factor`) significantly improved the model's ability to capture morning and evening commuter peaks.

- **Interaction Terms:** User behavior changes based on combinations of conditions. Interaction terms such as `working_temp` (working day $\times$ temperature) and `hum_wind` (humidity $\times$ windspeed) were introduced to capture nuanced behavioral shifts.

The baseline MLR equation took the form:

$$y_t = \beta_0 + \sum_{i=1}^k \beta_i X_{i,t} + \beta_{temp^2} temp_t^2 + \beta_{int}(workingday_t \times temp_t) + \epsilon_t \quad (1)$$

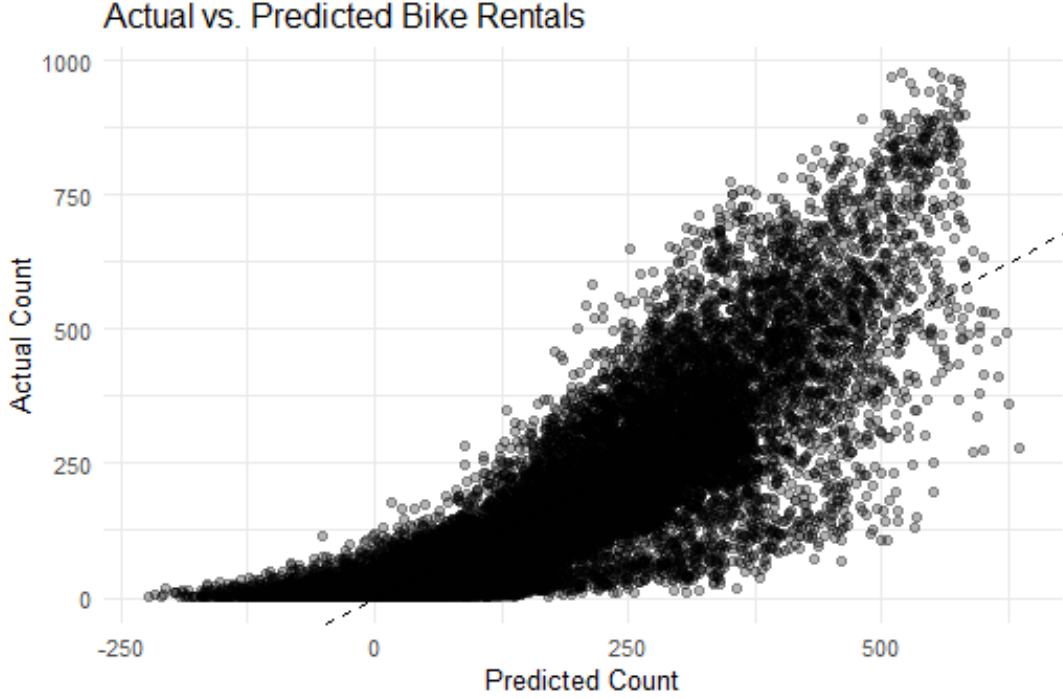


Figure 1: Actual vs. Predicted Bike Rentals from the baseline Multiple Linear Regression model in R.

To further understand the temporal nature of the dataset before applying time-series models, we analyzed the Autocorrelation and Partial Autocorrelation.

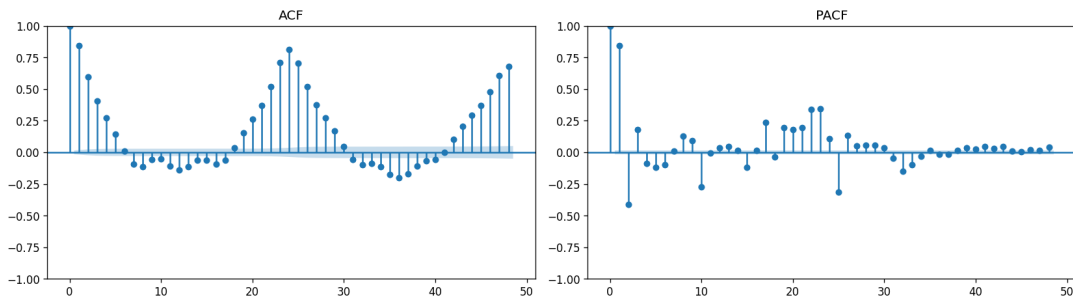


Figure 2: ACF and PACF plots demonstrating significant autocorrelation and daily (24-hour) seasonality.

3 Methodology and Model Selection

While linear models provide interpretability, time series data possesses inherent autocorrelation and complex non-linearities that MLR struggles to capture. Therefore, we transitioned to two advanced methodologies in Python.

3.1 Model 1: SARIMAX (Time Series Approach)

Rationale: As seen in the ACF plots, the bike rental data exhibits strong autocorrelation and distinct daily seasonality. A Seasonal Autoregressive Integrated Moving Average with eXogenous factors (SARIMAX) model natively models these temporal dependencies while explicitly incorporating our weather and calendar variables.

Formulation: The $\text{SARIMAX}(p, d, q) \times (P, D, Q, s)$ model is defined as:

$$\Phi_P(B^s)\phi_p(B)(1 - B^s)^D(1 - B)^d y_t = \beta X_t + \Theta_Q(B^s)\theta_q(B)\epsilon_t \quad (2)$$

Grid search was utilized to minimize the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), ensuring an optimal balance between model fit and complexity.

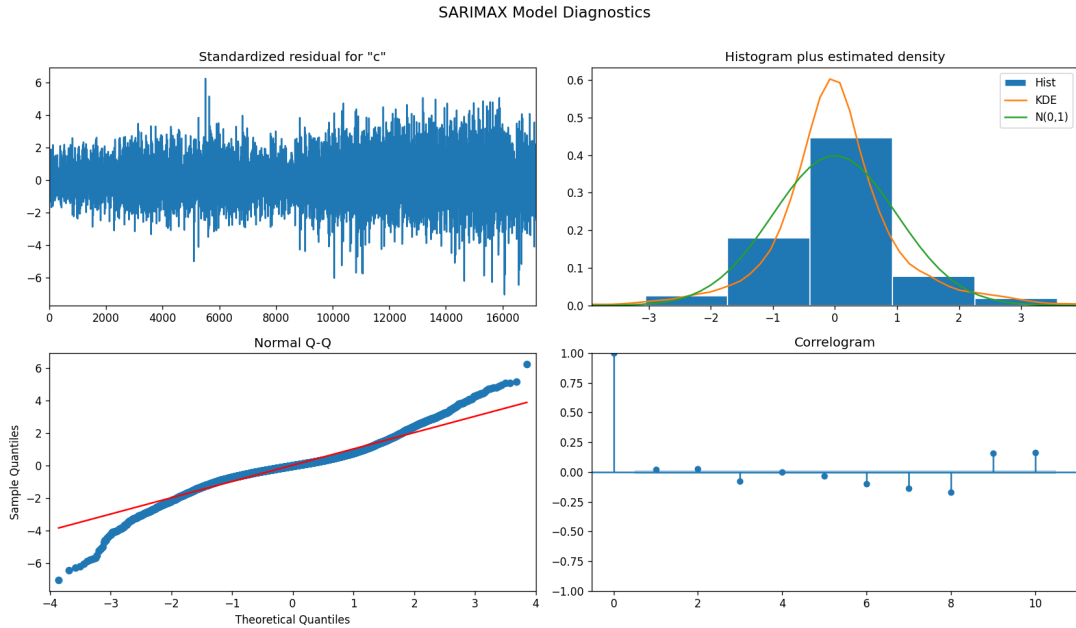


Figure 3: SARIMAX model diagnostics, illustrating residual distributions and ensuring white-noise properties are met.

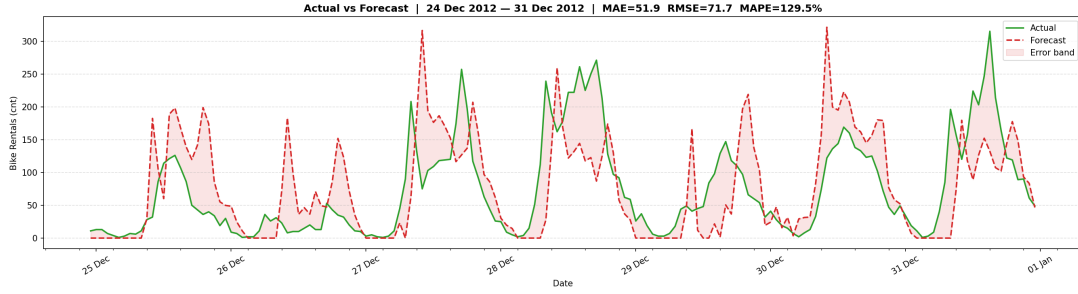


Figure 4: SARIMAX model forecasting performance against actual hourly demand.

3.2 Model 2: Histogram-based Gradient Boosting Regressor (Machine Learning Approach)

Rationale: Gradient Boosting captures highly non-linear interactions without needing manual specification. The `HistGradientBoostingRegressor` is highly efficient for larger datasets. To adapt this standard regression model for time-series forecasting, we engineered lag features ($t - 1$, $t - 24$) and rolling statistics to mimic the AR and seasonal components of SARIMAX.

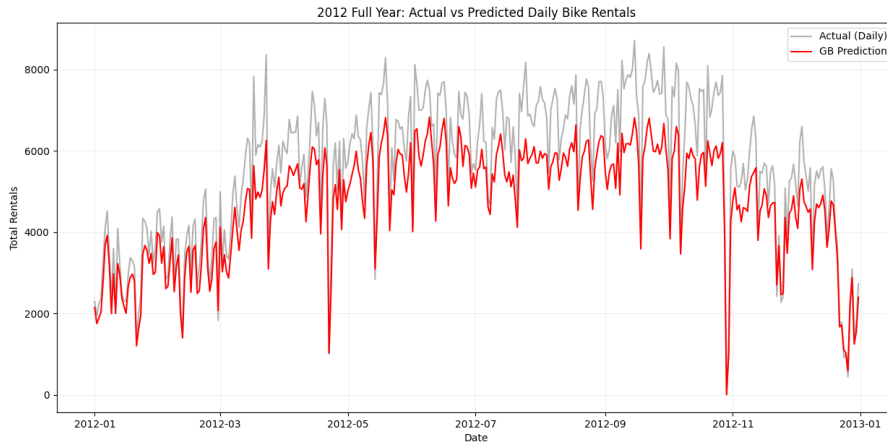


Figure 5: Daily aggregated forecasting performance of the Gradient Boosting model over a full year, capturing macro-level seasonal trends.

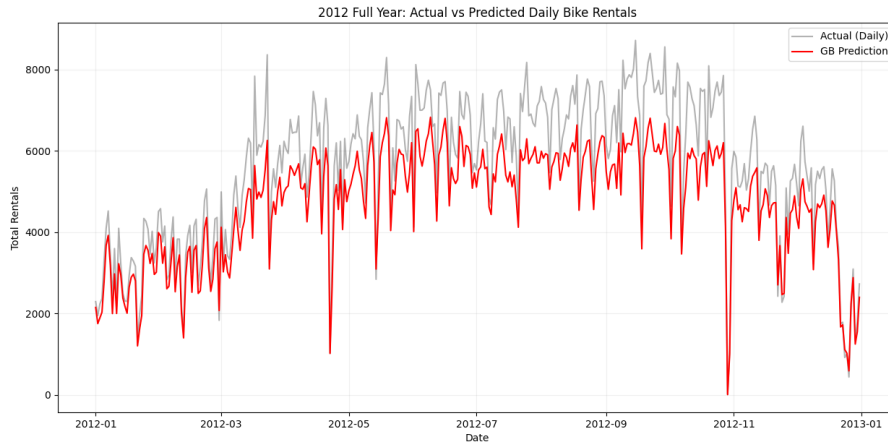


Figure 6: A zoomed-in, hourly snippet demonstrating the Gradient Boosting model’s ability to capture sharp commuter rush-hour peaks.

4 Final Comparison and Conclusion

The two models offer distinct trade-offs in forecasting bike demand:

1. **Handling of Relationships:** SARIMAX imposes a linear constraint on the exogenous variables alongside its temporal ARIMA structure. Conversely, Gradient Boosting natively learns complex, non-linear weather thresholds (e.g., steep drop-offs in rentals during extreme heat), making it often more accurate for sudden behavioral shifts.
2. **Feature Engineering vs. Native Temporal Modeling:** SARIMAX inherently understands time through its seasonal parameters. To achieve comparable temporal awareness in Gradient Boosting, we had to manually engineer lag and rolling window features.
3. **Interpretability and Diagnostics:** SARIMAX provides rigorous statistical diagnostics (Figure 3) to ensure white-noise residuals, making it highly interpretable from an inferential standpoint. Gradient Boosting functions more as a “black box,” focusing strictly on minimizing predictive error.

Conclusion: For pure predictive accuracy, especially during anomalous weather conditions or high-variance rush hours, the **Gradient Boosting** approach generally outperforms due to its non-linear flexibility. However, **SARIMAX** remains superior if the primary goal is statistical inference, understanding the exact marginal impact of specific exogenous variables, and providing confidence intervals backed by robust time-series theory.